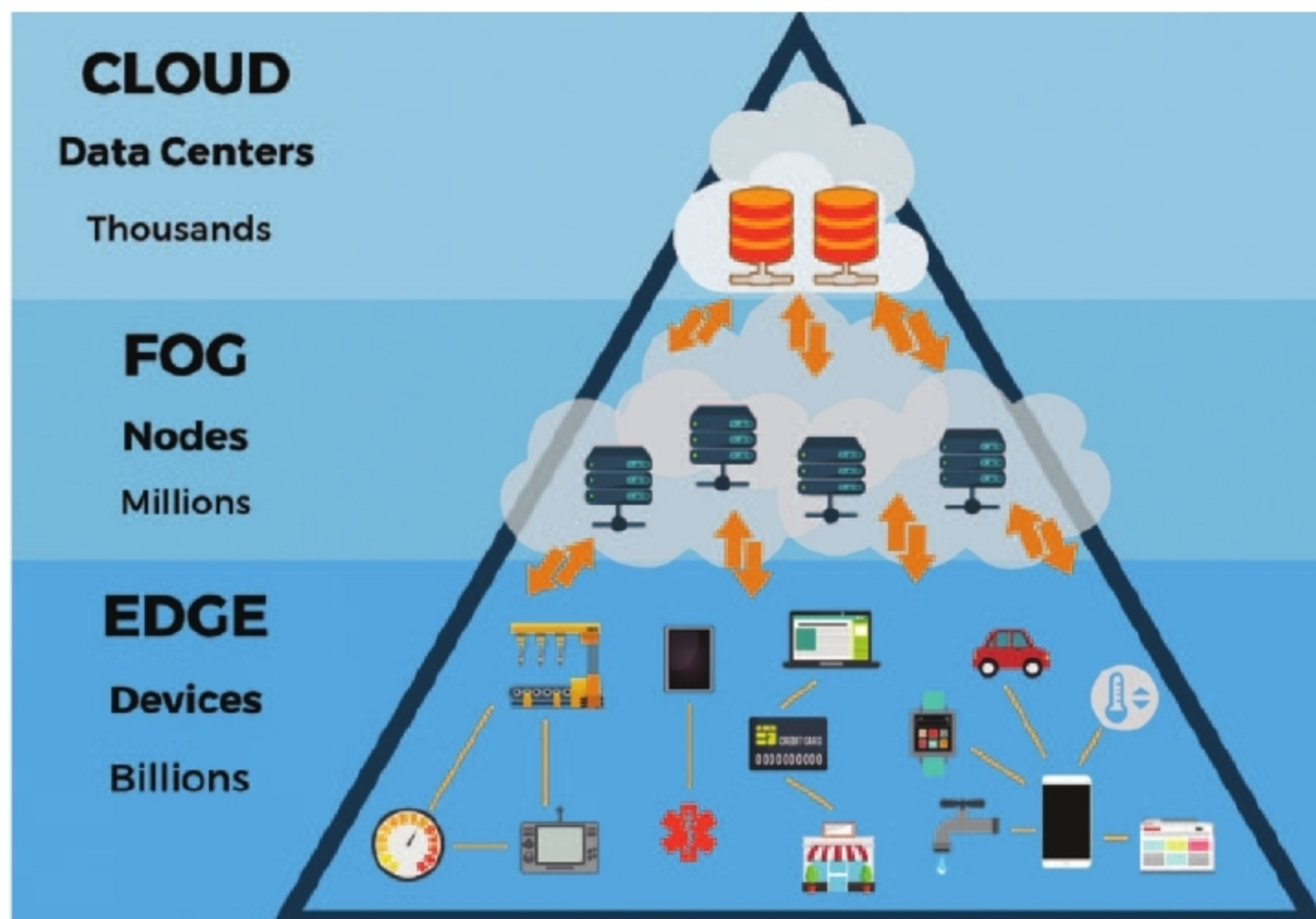


Edge and Fog Computing: Their Practical Uses

This article is based on a talk by Chetan Kumar S. at India Electronics Week 2019, organised by EFY Group, where he informed how Edge and Fog computing can be used to solve the problem of latency.



One popular application of Edge computing is city surveillance. Many surveillance cameras are coming up. For instance, Mumbai has thousands of CCTV cameras, and it plans to install many more. All these cameras will get traffic footage, which will help in identifying mobility problems and criminal activities. But it is practically impossible for security personnel to monitor them all. So, one option is to use computer vision and machine learning.

The machine learning process is related to data. In terms of camera surveillance, data from the cameras

needs to be collected and processed to perform various operations. With respect to bandwidth requirement, a single camera needs a speed of about 3Mbps for streaming an SD video and 5Mbps for an HD video. An SD video consumes 32GB data in a day, which means ten cameras will consume 1.7TB data per month. And with further increase in the number of cameras, data consumption will also increase.

Why we need Edge computing and where it can be used

While cost of the CPU and storage is one part of cloud computing, bandwidth is another significant part. Bandwidth

cost is not just in terms of pushing data to the cloud, it also includes cost of the servers and storage space.

If you want to avoid the issue of storing a large amount of data in the cloud to reduce cost, the best way to do that is to process it near the edge of the network, where data is being generated. Hence, the term Edge computing.

To automate this using computer vision and machine learning algorithms, a system is required rather than a human being to process data. For example, this way, data from the cameras at a traffic signal can be processed then and there itself.

